

Measuring Intervention-Conditioned Recoverability Profiles from a Fixed Post-Mismatch State

Same-State Counterfactual Evaluation for Physical AI Beyond the Average Case

Seonhye Gu

AI-Based Surgical Robot Innovation Lab

July 27, 2026

Abstract

Physical AI systems inevitably encounter situations in which the assumptions used for learning or planning no longer match the execution environment. Prior work has studied latent world representations, action-conditioned prediction, uncertainty-triggered handover, and recovery or backup policies. However, in the literature reviewed for this study, standardized evaluation that compares multiple intervention classes from the same mismatch-onset state remains limited.

We introduce a *same-state counterfactual evaluation* protocol that reconstructs a matched simulator execution state S immediately after a task-relevant mismatch and varies only the intervention applied after that state. In an Isaac Sim 4.1 and ORBIT-Surgical dual-arm reach proxy, we shift the target by 6 cm along the Y -axis and add a visibility-degradation proxy. We compare CONTINUE, REPLAN, REOBSERVE, RESHAPE, and HANDOVER responses. In a preregistered confirmatory cell with $n = 20$ seeds per branch, CONTINUE succeeded in 0/20 trials, whereas REPLAN succeeded in 19/20. REOBSERVE and RESHAPE succeeded in 17/20 and 18/20 trials, respectively. A visibility-threshold-based uncertainty rule selected HANDOVER in all 20 trials. Because the current HANDOVER branch does not implement human completion, terminal resolution after handover cannot yet be evaluated. A fixed situation rule selecting REOBSERVE succeeded in 17/20 trials.

These anchor-condition results support the hypothesis that a separate response-selection problem remains after mismatch detection. We do not claim a learned recoverability estimator, a learned world model, or clinical deployment. Instead, the study provides a Stage-1 measurement scaffold for future mismatch detectors, response selectors, informative-failure curricula, and clinically meaningful exception scenarios.

Keywords: Physical AI, recoverability, mismatch, counterfactual evaluation, replanning, surgical robotics, intervention selection

1 Introduction

1.1 Background

World-model research has shown that an agent can compress observations into latent states and predict how actions change future states. Ha and Schmidhuber [3] demonstrated that a visual encoder, a recurrent latent-dynamics model, and a compact controller can support policy learning inside a generated “dream” environment. Their VizDoom experiments also exposed a failure mode in which a policy exploits inaccuracies in the learned model, illustrating that success inside a model does not guarantee success in reality.

V-JEPA 2 extended this trajectory from latent prediction toward large-scale self-supervised video learning and real-robot planning. The model first learns an action-free representation from more than one

million hours of video and is then adapted with fewer than 62 hours of robot data to obtain the action-conditioned V-JEPA 2-AC model. The resulting system performs image-goal-based manipulation planning on Franka robots in previously unseen laboratory settings without environment-specific data collection or task-specific reward design [1].

These studies establish that internal representations and predictive dynamics can support physical action. They leave open a downstream question:

What should an embodied system do when its predictive or execution assumptions no longer match observed reality?

After a mismatch, a system may continue the current action sequence, update the plan, gather additional observations, reshape the environment, or request human assistance. Comparing such responses across different trajectories, however, confounds the effect of the response with differences in initial conditions and execution history.

1.2 Problem Definition

We compare multiple response classes from a matched simulator execution state S , reconstructed immediately after a mismatch using the same seed and nominal action replay.

Our central research question is:

At a fixed post-mismatch execution state S , do intervention classes yield separable recoverability profiles, and how does a multi-mode response menu differ from CONTINUE, a UQ-inspired binary rule, and a fixed situation rule in successful resolution and intervention-selection behavior?

Recoverability is treated here as an empirical outcome rather than a latent property or a learned estimate. For intervention a , onset state S , and intervention time t , we operationalize the recoverability profile as

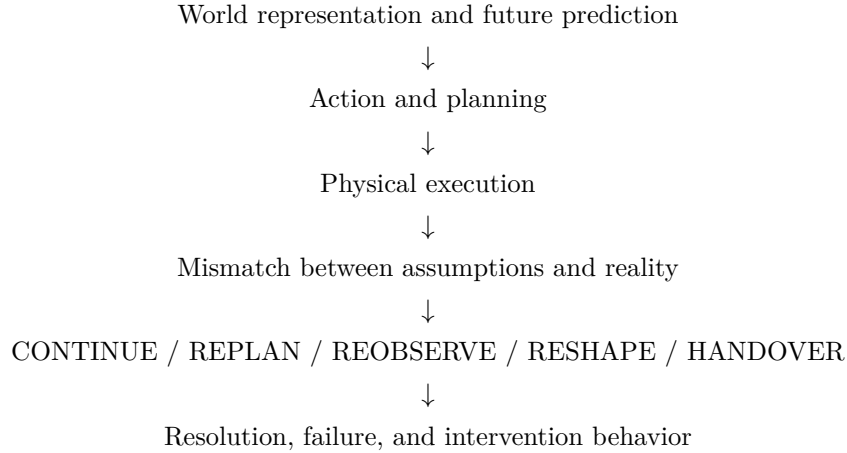
$$\hat{R}_a(S, t) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[y_i(S, a, t) = \text{success}], \quad (1)$$

where N is the number of trials in the response branch and y_i is the terminal outcome in trial i .

The study has four goals:

1. isolate the effect of intervention choice from differences in rollout history;
2. compare CONTINUE, REPLAN, REOBSERVE, RESHAPE, and HANDOVER within one response menu;
3. compare the menu with a binary uncertainty-triggered handover rule and a fixed situation rule; and
4. establish an empirical ruler for later learned response-selection models.

1.3 Program Storyline



Whereas *World Models* and V-JEPA 2 study prediction and planning through internal models, our study focuses on what happens *after* the current model or plan becomes inadequate for the observed execution state.

2 Related Work

2.1 World Models and Latent-Space Planning

World Models separates visual compression, recurrent latent dynamics, and control [3]. Its importance for this study lies not only in latent imagination but also in the model-exploitation example observed in VizDoom, where the controller takes advantage of inaccuracies in the learned simulator.

V-JEPA 2 predicts future features in a joint-embedding space rather than reconstructing pixels. Its action-conditioned extension, V-JEPA 2-AC, predicts future latent states from visual context and robot actions and uses image-goal planning to control a real robot [1]. The approach efficiently combines large-scale observation data with limited interaction data. However, its main objective is latent physical planning, not explicit interpretation of prediction–observation mismatch or comparison among multiple post-mismatch intervention modes.

Our work therefore does not propose a new world model. It studies an intervention-evaluation layer that may sit on top of world-model-based or other Physical AI systems.

2.2 Uncertainty and Human Intervention in Surgical Robotics

Surgical robot-learning work has investigated whether uncertainty can provide an early indication of failure under changes in tissue geometry, stiffness, or visual conditions. Thompson et al. [7] compare deep ensembles and Monte Carlo dropout and use uncertainty to decide when a robot should request human intervention.

Agreement Volatility further studies temporal changes in model agreement to determine when to hand control to a person and when to resume autonomy [6]. These methods connect uncertainty estimation with handover decisions. Their action space is largely binary: continue autonomously or escalate to a human. Our protocol instead evaluates a broader menu that includes REPLAN, REOBSERVE, and RESHAPE before or alongside HANDOVER.

2.3 Situation Handling, Replanning, and Recovery

Robots operating in changing environments require mechanisms for handling situations that were not captured in the initial plan. VAP-TAMP combines active perception, vision-language-based situation assessment, scene representation, and task-and-motion planning to handle execution-time changes [5].

Runtime-assurance and recovery-policy work similarly considers switching from a task policy to a backup or safety policy when continued execution becomes unsafe or ineffective. Such architectures are important for long-horizon autonomy, but they do not primarily standardize parallel evaluation of a multi-mode response menu from a matched mismatch-onset state.

Our contribution is an evaluation protocol rather than a new planning architecture.

2.4 Environment Reshaping and Shared Autonomy

MEDiC automates tissue exposure and reshaping by combining differentiable physics, perceptual feedback, and deformable Jacobian analysis [4]. In that work, reshaping is a central surgical function. In our protocol, RESHAPE is treated as one timed intervention among several possible responses after mismatch.

2.5 Positioning

Table 1: Positioning relative to adjacent research lines.

Research line	Primary focus	Role in this work
World models	Internal representation and prediction	Evaluate what follows when assumptions break
V-JEPA family	Latent physical planning	Compare post-mismatch responses
Surgical UQ	Failure detection and handover	Expand from binary handover to a multi-mode menu
Recovery / RTA	Task-to-backup switching	Compare interventions from matched onset states
VAP-TAMP	Situation understanding and replanning	Evaluate response profiles, not architecture
MEDiC	Tissue reshaping for exposure	Treat RESHAPE as one intervention option

The key distinction is the reconstruction of a matched mismatch-onset state that satisfies predefined simulator-state and replay-QC criteria, followed by intervention-only branching and terminal-resolution comparison.

3 Method

3.1 Experimental Environment

The controller is scripted inverse kinematics rather than a learned RL policy. This design avoids conflating policy-learning performance with intervention evaluation in the Stage-1 study.

3.2 Same-State Counterfactual Evaluation

The protocol proceeds as follows:

1. fix the random seed;
2. execute the nominal rollout for steps 0–19 and record the actions;

Table 2: Experimental environment.

Component	Setting
Simulator	Isaac Sim 4.1.0
Framework	Isaac Lab 1.0.0, ORBIT-Surgical
Task	Isaac-Reach-Dual-STAR-IK-Rel-Play-v0
Controller	Scripted 6D IK-Rel, body 13
Episode horizon	160 steps
Episode length	20 seconds
D0 execution	RunPod, RTX 4090
D1–D3 execution	VESSL, A100
Confirmatory sample	Seeds 0–19 per branch, $n = 20$

3. at step 20, shift the target by 6 cm along the Y -axis;
4. in D0, apply occlusion level L1 with visibility set to 0.35;
5. reset each response branch with the same seed;
6. replay the recorded nominal actions to reconstruct state S ;
7. vary only the post- S response; and
8. evaluate the terminal outcome.

Nominal rollout $\rightarrow$ Mismatch at $S \rightarrow$ Replay $\rightarrow$ Response fork $\rightarrow$ Terminal judge. (2)

All 100 branches in D0 and all 80 branches in D1–D3 passed the predefined replay-QC criterion. In this paper, “same-state” does not imply that every hidden simulator variable is byte-identical. It denotes a matched execution state reconstructed by the same seed and nominal action replay and verified by the study’s QC checks.

3.3 Response Classes

CONTINUE. Maintain the pre-mismatch target or execution assumption and continue the current behavior.

REPLAN_{d20}. Hold for 20 steps and then update the scripted IK target using the shifted goal. This is a goal-conditioned IK target update rather than a general task-and-motion planning system.

REOBSERVE. Hold, restore visibility, and then update the target.

RESHAPE. Apply an environment-reshaping proxy and then update the target.

HANDOVER. Transition to a human-assistance stub. Human completion is not implemented in the current study, so post-handover terminal resolution cannot be evaluated.

3.4 Rule Baselines

B2: UQ-Inspired Binary Rule. A visibility- or distance-based threshold selects HANDOVER. In the anchor condition, the visibility rule fires in all trials.

B3: Situation Rule. A fixed mapping selects the intervention:

occluded shift $\rightarrow$ REOBSERVE, (3)

shift only $\rightarrow$ REPLAN. (4)

3.5 Success Criterion

A trial is labeled a successful automatic resolution when all of the following conditions hold:

1. the end-effector is within 0.02 m of the shifted target;
2. no forbidden axis-aligned bounding box is entered; and
3. the branch does not terminate in the HANDOVER stub.

Occlusion is currently implemented as a controller-gain proxy rather than a geometric mesh or RGB occluder. The results therefore should not be generalized directly to visual occlusion in clinical scenes.

4 Results

4.1 D0: Multi-Mode Profile at the Anchor State

Table 3: D0 multi-mode recoverability profile.

Response	Successes	Empirical success rate
CONTINUE	0/20	0%
REPLAN _{d20}	19/20	95%
REOBSERVE	17/20	85%
RESHAPE	18/20	90%
HANDOVER	–	Not evaluable

A large difference was observed between CONTINUE and REPLAN_{d20}. REOBSERVE and RESHAPE also provided successful resolution paths. These results indicate that recoverability in the anchor condition is better represented as an intervention-conditioned profile than as a single binary state attribute.

4.2 D1: No-Occlusion Control

Table 4: D1 no-occlusion control.

Response	Successes
CONTINUE	1/20
REPLAN _{d20}	19/20

REPLAN_{d20} remained at 19/20, whereas CONTINUE increased from 0/20 to 1/20. Within this narrow condition, failure to update the goal after the shift appears more consequential than the presence or absence of the gain-scale occlusion proxy. This result does not support a general claim that occlusion is unimportant.

4.3 D2: UQ-Inspired Binary Rule

Table 5: D2 UQ-inspired binary rule.

Outcome	Count
HANDOVER selected	20/20
Automatic terminal resolution	Not evaluable

The threshold rule maps every anchor-condition trial to HANOVER. Because human completion is not implemented, this result should not be interpreted as evidence that human intervention is ineffective. It shows only that the current binary rule does not distinguish among alternative non-handover responses.

4.4 D3: Situation Rule

The fixed situation rule selects REOBSERVE in the occluded-shift condition and succeeds in 17/20 trials.

* Human completion after HANOVER is not implemented.

Table 6: Response menu and rule-baseline comparison.

Comparator	Selected response	Automatic terminal resolution
Best observed menu response	REPLAN _{d20}	19/20 (95%)
B2 UQ-inspired rule	HANOVER in 20/20	Not evaluable*
B3 situation rule	REOBSERVE	17/20 (85%)

The best observed response in the menu has a higher empirical success proportion than the B3 situation rule. B2 cannot be compared directly on terminal resolution because its selected branch lacks human completion. The best-of-menu value is an oracle-style post hoc comparison, not the output of a learned online selector.

4.5 Study 002: Exploratory Informative-Failure Generation

Study 002 is kept separate from the main Paper-001 contribution. Its objective is to generate and rank perturbation specifications that satisfy

$$\text{CONTINUE fails} \wedge \text{REPLAN succeeds.} \quad (5)$$

In Phase 1, Gaussian sampling showed higher yield, while diffusion sampling showed higher diversity. The per-specification mock-to-Isaac rank correlation was low ($\rho = 0.15$). In Phase 2, after aligning the occlusion definition and protocol over a 20-specification pack, the top mock-ranked tier achieved an Isaac informative rate of 1.0, the bottom tier achieved 0.3, and the rank correlation increased to $\rho = 0.899$.

These findings suggest that mock ranking may support simulator-scenario selection when the mock-to-simulator contract is aligned. They do not establish a production-level dreamer or a learned world model.

5 Discussion

5.1 Detection and Response Selection Should Be Evaluated Separately

The B2 rule detects the visibility condition and selects HANOVER in all trials. In contrast, REPLAN_{d20}, REOBSERVE, and RESHAPE provide high automatic-resolution rates from the same matched state.

$$m_t = D(\hat{s}_{t+1}, s_{t+1}, c_t), \quad (6)$$

$$a_t^R = \pi_R(m_t, s_t, g), \quad (7)$$

where D is a mismatch detector and π_R is a response selector. We do not train either component in this study. The empirical results motivate evaluating them as distinct problems.

5.2 Recoverability Is Intervention-Conditioned

The same matched state is not resolved by CONTINUE but is resolved in most REPLAN, REOBSERVE, and RESHAPE branches. Thus, labeling a state as simply “recoverable” or “unrecoverable” is insufficient.

$$R(S) \rightarrow \{R_{\text{continue}}(S), R_{\text{replan}}(S), R_{\text{reobserve}}(S), R_{\text{reshape}}(S), R_{\text{handover}}(S)\}. \quad (8)$$

Recoverability depends on the available intervention set and, potentially, on intervention timing.

5.3 Connection to World-Model Research

World models and V-JEPA-style predictive representations support action selection by simulating or scoring future states [1, 3]. Yet any learned or engineered model remains incomplete. Our study asks four downstream questions:

1. Is the gap between predicted and observed state a meaningful mismatch?
2. Which intervention classes preserve resolution from that state?
3. When should the system replan, reobserve, reshape the environment, or request assistance?
4. Which mismatch events should be retained as evidence for later model revision?

The study therefore occupies the prediction–execution interface: it evaluates responses and outcomes after a mismatch rather than replacing the underlying world model.

5.4 Relation to V-JEPA 2.1

V-JEPA 2.1 further improves dense feature learning and real-robot grasping performance [2]. These advances strengthen the case for predictive representations in embodied control. Explicit mismatch-conditioned response comparison and intervention-profile measurement remain separate problems.

5.5 Toward Physical AI Beyond the Average Case

The current experiments use a simplified reach proxy and do not directly support claims about disability, anatomical diversity, or underserved populations. The longer-term program is more modestly framed: instead of treating every deviation from training assumptions as noise to be removed, it asks when an exception should change the system’s response and eventually its model of the world.

Stage 1 asks:

When execution assumptions break, which response preserves successful resolution?

Later stages may ask:

Which exceptions are transient noise, and which provide evidence that the system’s representation or assumptions should be revised?

6 Limitations

1. The study uses a single surgical-reach proxy and one primary anchor cell.
2. Occlusion is represented by a gain-scale proxy, not geometric or RGB occlusion.
3. The controller is scripted IK rather than a learned policy or world model.
4. HANDOVER is a stub without human completion.
5. Best-of-menu is an oracle-style post hoc comparison, not an online selector.
6. Each branch contains only $n = 20$ trials.
7. The current environment omits tissue deformation, bleeding, smoke, anatomical variation, contact force, and clinical workflow.

7 Conclusion

We introduced a same-state counterfactual evaluation protocol for comparing multiple post-mismatch interventions from a matched simulator execution state. In the 6 cm shift plus L1 occlusion anchor condition, CONTINUE succeeded in 0/20 trials and REPLAN_{d20} in 19/20. REOBSERVE and RESHAPE succeeded in 17/20 and 18/20 trials. A UQ-inspired binary rule selected HANDOVER in all trials, but terminal resolution after handover could not be evaluated because human completion was not implemented.

The results support two working hypotheses:

1. a separate response-selection problem remains after mismatch detection; and
2. recoverability should be measured as an intervention-conditioned profile rather than a single state property.

We do not claim a learned world model, a recoverability estimator, or clinical autonomy. The contribution is a Stage-1 measurement scaffold for later mismatch detectors, response selectors, informative-failure curricula, and clinically grounded exception scenarios.

8 Future Work

Multi-cell generalization. Evaluate whether intervention-profile ordering persists across severity and occlusion levels.

Geometric occlusion. Replace the gain-scale proxy with mesh- or RGB-based occlusion.

Timing study. Design a dedicated intervention-delay experiment because the current smoke grid did not show a clear timing cliff.

Learned selector. Train a model to select a response mode from state S , mismatch features, and uncertainty signals.

Human handover completion. Implement simulated or real operator completion to measure both terminal resolution and intervention burden.

Surgical validity. Extend the scenario set to tissue deformation, tool slip, occlusion, bleeding, and goal changes.

Mock-to-Isaac contract. Test whether mock ranking continues to predict Isaac informative rate without manual protocol alignment.

References

- [1] Mahmoud Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, et al. V-jepa 2: Self-supervised video models enable understanding, prediction and planning. *arXiv preprint arXiv:2506.09985*, 2025. URL <https://arxiv.org/abs/2506.09985>.
- [2] Mahmoud Assran et al. V-jepa 2.1: Unlocking dense features in video self-supervised learning. *arXiv preprint arXiv:2603.14482*, 2026. URL <https://arxiv.org/abs/2603.14482>.
- [3] David Ha and Jürgen Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2018. URL <https://arxiv.org/abs/1803.10122>.

- [4] Xinyang Liang, Chien-Pin Wang, Nikhil U. Shinde, Fang Liu, Frank Richter, and Michael Yip. Medic: Autonomous surgical robotic assistance to maximizing exposure for dissection and cautery. In *Proceedings of the IEEE International Conference on Robotics and Automation*, pages 6889–6895, 2025. URL <https://arxiv.org/abs/2409.14287>.
- [5] A. Oloo et al. Robot planning and situation handling with active perception. *arXiv preprint arXiv:2604.26988*, 2026. URL <https://arxiv.org/abs/2604.26988>.
- [6] Jacob Thompson, Benjamin Jordan, Daniel S. Brown, and Alan Kuntz. Agreement volatility: A second-order metric for uncertainty quantification in surgical robot learning. In *Proceedings of the 9th Conference on Robot Learning*, pages 3685–3703, 2025. URL <https://proceedings.mlr.press/v305/thompson25a.html>.
- [7] Jacob Thompson, Ryan Koe, An Le, Gregory Goodman, Daniel S. Brown, and Alan Kuntz. Early failure detection in autonomous surgical soft-tissue manipulation via uncertainty quantification. *arXiv preprint arXiv:2501.10561*, 2025. URL <https://arxiv.org/abs/2501.10561>.